Observe that:

Let $f:[a,b]\to\mathbb{R}$, $n\in\mathbb{N}$, $\alpha,\beta\in[a,b]$. Suppose $f^{(n-1)}$ exists on $[a,b]$. Define
$$Q(t)=\frac{f(t)-f(\beta)}{t-\beta}\quad(t\neq\beta).$$

Then
$$f(\beta)=\sum_{k=0}^{n-1}\frac{f^{(k)}(\alpha)}{k!}\cdot(\beta-\alpha)^k+\frac{Q^{(n-1)}(\alpha)}{(n-1)!}\cdot(\beta-\alpha)^n$$

Proof. Derivating $(n-1)$ times to:
$$(t-\beta)\cdot Q(t)=f(t)-f(\beta)$$

Then,
$$Q(t)+(t-\beta)Q'(t)=f'(t)$$
$$Q'(t)+Q'(t)+(t-\beta)Q''(t)=f''(t)$$
$$\vdots$$
$$(n-1)\cdot Q^{(n-2)}(t)+(t-\beta)\cdot Q^{(n-1)}(t)=f^{(n-1)}(t)$$

Now,
$$\begin{aligned}f(\beta)&=f(t)+Q(t)\cdot(\beta-t)\\&=f(t)+f'(t)\cdot(\beta-t)+Q'(t)(\beta-t)^2\\&=f(t)+f'(t)(\beta-t)+\frac{f''(t)}{2!}(\beta-t)^2+\frac{Q''(t)}{2!}(\beta-t)^3\\&\quad\vdots\\&=\sum_{k=0}^{n-1}\frac{f^{(k)}(t)}{k!}\cdot(\beta-t)^k+\frac{Q^{(n-1)}(t)}{(n-1)!}\cdot(\beta-t)^n.\end{aligned}$$

$$g(t)=(\beta-t)^n$$

Consider $F(t)=\sum_{k=0}^{n-1}\frac{Q^{(k)}(t)}{k!}(\beta-t)^k$ and $G(t)=\sum_{k=0}^{n-1}\frac{g^{(k)}(t)}{k!}(\beta-t)^k$

Then,
$$\begin{aligned}F'(t)=\;&Q'(t)-Q^{(1)}(t)+Q^{(2)}(t)(\beta-t)\\&-\frac{1}{2!}\cdot2\cdot Q^{(2)}(t)\cdot(\beta-t)+\frac{1}{2!}\cdot Q^{(3)}(t)\cdot(\beta-t)^2\\&-\frac{1}{3!}\cdot3\cdot Q^{(3)}(t)(\beta-t)^2+\frac{1}{3!}Q^{(4)}(t)\cdot(\beta-t)^3\\&\quad\vdots\\&-\frac{1}{(n-1)!}\cdot(n-1)\cdot Q^{(n-1)}(t)(\beta-t)^{n-2}+\frac{1}{(n-1)!}\cdot Q^{(n)}(t)(\beta-t)^{n-1}\\=\;&\frac{Q^{(n)}(t)}{(n-1)!}\cdot(\beta-t)^{n-1}\end{aligned}$$

Now, by the Generalized Mean-Value Theorem,
$$F'(s)\cdot[F(\alpha)-F(\beta)]=G'(s)[G(\alpha)-G(\beta)]$$

Let f be a map.

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(\alpha)}{k!}(x-\alpha)^k = f(\alpha) + \frac{f'(\alpha)}{1!}(x-\alpha) + \frac{f''(\alpha)}{2!}(x-\alpha)^2 + \cdots + \frac{f^{(n)}(\alpha)}{n!}(x-\alpha)^n.$$

$P_n(\alpha) = f(\alpha)$

$P_n'(\alpha) = f'(\alpha)$

$\vdots$

$P_n^{(n)}(\alpha) = f^{(n)}(\alpha).$

$P_n^{(n+1)}(\alpha) = 0$

$\vdots$

Let $g_n(x) = (x-\alpha)^{n+1}$. Then